Jeonghwan Kim

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Research Interests

Multimodal Foundation Models & Representation Learning

Vision-Language Models (VLMs), Vision-Language-Action (VLA) Models, World Action Models (WAMs), Multimodal Representation Learning, Visual Tokenizers, Retrieval-Augmented Generation (RAG), Multimodal Post-training, Model Interpretability and Robustness.

Education

University of Illinois at Urbana-Champaign (UIUC) — Champaign, USA
Ph.D. Candidate in Computer Science (Advisor: Heng Ji) — Aug 2023 – Present

KAIST — Daejeon, South Korea
M.S. in Computer Science (Advisor: Sung-Hyon Myaeng) — Feb 2022
Thesis Committee: Sung-Hyon Myaeng, Alice Oh, Junho Lim
GPA: 4.13/4.30

Handong Global University — Pohang, South Korea
B.S. in Computer Science & Electrical Eng. (Advisor: Heeyoul Henry Choi) — Feb 2020
Magna Cum Laude (Overall GPA: 4.11/4.5 | Major GPA: 4.21/4.5)

Experience

Meta — *Research Scientist Intern* — Redmond, USA
Developed language-guided world action models (WAM) for smart glasses. — May 2026 – Aug 2026

Meta — *Research Scientist Intern (Part-time Student Researcher)* — Redmond, USA
Developed end-to-end pixel-level RAG policy learning for multimodal LLMs. — May 2025 – Oct 2025

Amazon — *Applied Scientist Intern* — Bellevue, USA
Developed a generalizable multimodal encoder for geospatial applications. — May 2024 – Aug 2024

UIUC — *Graduate Research Assistant* — Champaign, USA
Research on Multimodal LLMs and reasoning. — Aug 2023 – Present

KAIST — *Researcher* — Daejeon, Korea
Developed graph and RAG-based language modeling approaches for QA. — Mar 2022 – Jul 2023

Republic of Korea Marine Corps — *Sergeant (Honorably Discharged)* — Pohang, Korea
Served as a rifleman and interpreter for the ROK-US Combined Marine Command. — Mar 2015 – Dec 2016

Publications

'*' indicates equal contribution and co-first author

[1] **Mixture of Layers: Dynamic Layer Routing for Visual Reasoning**
Jeonghwan Kim*, *Sofia Stoica**, *Jiwan Chung, Ansel Blume, Hyeonjeong Ha, Zhenhailong Wang, Xin Luna Dong, Heng Ji*
NeurIPS 2026

[2] **Advancing Creative Physical Intelligence in Large Multimodal Models**
*Cheng Qian, Hyeonjeong Ha, Jiayu Liu, **Jeonghwan Kim**, Emre Can Acikgoz, Bingxuan Li, Kunlun Zhu, Jiateng Liu, Aditi Tiwari, Zhenhailong Wang, Xiusi Chen and Heng Ji*

NeurIPS 2026

[3] **PixSearch: Region-Grounded Retrieval for Knowledgeable Large Multimodal Models**
Jeonghwan Kim*, Renjie Tao, Sanat Sharma, Jiaqi Wang, Kai Sun, Zhaojiang Lin, Seungwhan Moon, Lambert Mathias, Anuj Kumar, Heng Ji, Xin Luna Dong*

NeurIPS 2026 VLM4RWD (Oral)

[4] **PARTONOMY: Large Multimodal Models with Part-Level Visual Understanding**
Jeonghwan Kim*, Ansel Blume*, Hyeonjeong Ha, Elen Chatikyan, Xiaomeng Jin, Khanh Duy Nguyen, Nanyun Peng, Kai-Wei Chang, Derek Hoiem and Heng Ji*

NeurIPS 2025 (Spotlight)

[5] **ProMac: Language-Guided World Action Modeling for Single-View Egocentric Navigation**
Jeonghwan Kim*, Maxim Arap, Marios Galanis, Hyeonjeong Ha, Junhyeok Kim, Anuj Kumar, Xin Luna Dong, Heng Ji, Seungwhan Moon*

Preprint (Under Review)

[6] **Generalizable VLA Finetuning via Representation Anchoring and Language-Action Alignment**
*Dwip Dalal, Shivansh Patel, Chahit Jain, **Jeonghwan Kim**, Utkarsh Mishra, Alex Baratian, Hyeonjeong Ha, Heng Ji, Svetlana Lazebnik, Unnat Jain*

Preprint (Under Review)

[7] **Jury Duty: Calibration and Orientation Failures in MLLM-as-a-Judge Under Cultural Ambiguity**
*Daniel Lee, Harshit Sharma, Eunkyu Park, Pranav Narayanan Venkit, **Jeonghwan Kim**, Kah Mun Chia, Andreas Vlachos, Shafiq Joty*

EMNLP 2026

[8] **MemGuard: Preventing Memory Contamination in Long-Term Memory-Augmented Large Language Models**
*Hyeonjeong Ha, **Jeonghwan Kim**, Cheng Qian, Jiayu Liu, William M. Campbell, Yue Wu, Yuji Zhang, Kathleen McKeown, Dilek Hakkani-Tür, Heng Ji*

EMNLP 2026, Findings

[9] **Constructive Distortion: Improving MLLMs with Attention-Guided Image Warping**
*Dwip Dalal, Gautam Vashishtha, Utkarsh Mishra, **Jeonghwan Kim**, Madhav Kanda, Hyeonjeong Ha, Svetlana Lazebnik, Heng Ji, Unnat Jain*

ICLR 2026

[10] **PEARL: Self-Evolving Assistant for Time Management with Reinforcement Learning**
*Bingxuan Li, **Jeonghwan Kim**, Cheng Qian, Xiusi Chen, Eitan Anzenberg, Niran Kundapur, Heng Ji*

ACL 2026

[11] **MM-PoisonRAG: Disrupting Multimodal RAG with Local and Global Knowledge Poisoning Attacks**
Hyeonjeong Ha, Qiusi Zhan*, **Jeonghwan Kim**, Dimitrios Bralios, Saikrishna sanniboina, Nanyun Peng, Kai-Wei Chang, Daniel Kang, Heng Ji*

ACL 2026

[12] **Infogent: An Agent-based Framework for Web Information Aggregation**
Jeonghwan Kim*, Revanth Gangi Reddy*, Sagnik Mukherjee*, Zhenhailong Wang*, Dilek Hakkani-Tur, Heng Ji*

NAACL 2025, Findings

[13] **SYNTHIA: Novel Concept Design with Affordance Composition**
*Hyeonjeong Ha, Xiaomeng Jin, **Jeonghwan Kim**, Jiateng Liu, Zhenhailong Wang, Khanh Duy Nguyen, Ansel Blume, Nanyun Peng, Kai-Wei Chang, Heng Ji*

ACL 2025

[14] **Search and Detect: Training-Free Long Tail Object Detection via Web-Image Retrieval**
*Mankeerat Sidhu, Hetarth Chopra, Ansel Blume, **Jeonghwan Kim**, Revanth Gangi Reddy, Heng Ji*

CVPR 2025

[15] **Aligning LLMs with Individual Preferences via Interaction**
*Shujin Wu, May Fung, Cheng Qian, **Jeonghwan Kim**, Dilek Hakkani-Tur, Heng Ji*

COLING 2025

[16] **Finer: Investigating and Enhancing Fine-Grained Visual Concept Recognition in Large Vision Language Models**
Jeonghwan Kim*, Heng Ji*
EMNLP 2024

[17] **Why So Gullible? Enhancing the Robustness of Retrieval-Augmented Models against Counterfactual Noise**
Jeonghwan Kim*, Giwon Hong*, Junmo Kang*, Sung-Hyon Myaeng, Joyce Jiyoung Whang*
NAACL 2024, Findings

[18] **ARMADA: Attribute-Based Multimodal Data Augmentation**
*Xiaomeng Jin, **Jeonghwan Kim**, Yu Zhou, Kuan-Hao Huang, Te-Lin Wu, Nanyun Peng, Heng Ji*
Advancing Natural Language Process for Wikipedia@EMNLP 2024

[19] **MIRACLE: An Online, Explainable Multimodal Interactive Concept Learning System**
Ansel Blume, Khanh Duy Nguyen*, Zhenhailong Wang, Yangyi Chen, Michal Shlapentokh-Rothman, Xiaomeng Jin, **Jeonghwan Kim**, Zhen Zhu, Jiateng Liu, Kuan-Hao Huang, Mankeerat Sidhu, Xuanming Zhang, Vivian Liu, Raunak Sinha, Te-Lin Wu, Abhay Zala, Elias Stengel-Eskin, Da Yin, Yao Xiao, Utkarsh Mall, Zhou Yu, Kai-Wei Chang, Camille Cobb, Karrie Karahalios, Lydia Chilton, Mohit Bansal, Nanyun Peng, Carl Vondrick, Derek Hoiem, Heng Ji*
ACM MM Technical Demos, 2024

[20] **FinePrompt: Unveiling the Role of Finetuned Inductive Bias on Compositional Reasoning in GPT-4**
Jeonghwan Kim*, Giwon Hong*, Sung-Hyon Myaeng, Joyce Jiyoung Whang*
EMNLP 2023, Findings

[21] **Graph-Induced Transformers for Efficient Multi-Hop Question Answering**
*Giwon Hong, **Jeonghwan Kim**, Junmo Kang, Sung-Hyon Myaeng*
EMNLP 2022

[22] **Exploiting Numerical-Contextual Knowledge to Improve Numerical Reasoning in Question Answering**
Jeonghwan Kim*, Junmo Kang, Kyung-min Kim, Giwon Hong, Sung-Hyon Myaeng*
NAACL 2022, Findings

[23] **Have You Seen That Number? Investigating Extrapolation in Question Answering Models**
Jeonghwan Kim*, Giwon Hong, Kyung-min Kim, Junmo Kang, Sung-Hyon Myaeng*
EMNLP 2021

[24] **Leveraging Order-Free Tag Relations for Context-Aware Recommendation**
*Junmo Kang, **Jeonghwan Kim**, Suwon Shin, Sung-Hyon Myaeng*
EMNLP 2021

[25] **Can You Distinguish Truthful from Fake Reviews? User Analysis and Assistance Tool for Fake Review Detection**
Jeonghwan Kim*, Junmo Kang*, Suwon Shin*, Sung-Hyon Myaeng*
HCI+NLP Workshop, EACL 2021

[26] **Maximizing Efficiency of Language Model Pre-training for Learning Representation**
Jeonghwan Kim*, Junmo Kang*, Suwon Shin*, Jaeyoung Jo*, Sung-Hyon Myaeng*
KSC 2020

[27] **Object Classification on Raw Radar Data Using Convolutional Neural Networks**
*Heejae Han, **Jeonghwan Kim**, Junyoung Park, YuJin Lee, Hyunwoo Jo, Yonghyeon Park, Eric T Matson, Seongha Park*
IEEE SAS 2019

[28] **Towards the Development and Realization of an Undetectable Stealth UAV**
*Jiyeon Oh, Daeun Choe, Chanhui Yun, **Jeonghwan Kim**, Michael Hopmeier*
IEEE IRC 2019

Academic Services

Reviewer - ICLR 2027, AAAI 2027, TMLR, NeurIPS 2026, ICML 2026, ECCV 2026, CVPR 2026, ACL 2026, EACL 2026, SEA@NeurIPS 2025, AI4Research@AAAI 2025, EMNLP 2025, ACL 2025, ACL-SRW 2025, AAAI 2025, ACL 2024, Language + Molecules@ACL 2024, KnowledgeLM@ACL 2024, EMNLP 2024, EMNLP 2023

Honors & Awards

- **2026: *Capital One Ph.D. Fellowship***: Capital One Center for Generative AI Safety, Knowledge Systems and Cybersecurity (ASKS) @ UIUC
- **2026: *Gold Reviewer - ICML, 2026*** (International Conference on Machine Learning)
- **2026: *Great Reviewer - ACL Rolling Review***, January, 2026 (Association for Computational Linguistics)
- **2025: *NeurIPS 2025 Spotlight Publication*** - PARTONOMY: Large Multimodal Models with Part-Level Visual Understanding
- **2020: *Graduated with Honors in CSE, Magna Cum Laude*** (Handong Global University)
- **2020: *Finalist (Top 3)***, COMEUP2020 AI CHAMPIONSHIP (K-Startup)
- **2019: *Grand Prize***, Capstone Graduation Project Festival (Handong Global University, CSEE)
- **2018: *Certificate of Merit***, IITP Capstone 9 Program (Purdue University) - Selected among 30 students across South Korea
- **2018: *Finalist (Top 15)***, NAVER AI Hackathon (NAVER AI RUSH)
- **2017: *National Science and Technology Scholarship***, Korea Student Aid Foundation (KOSAF)

Invited Talks

- **Sep 2026: *Australasin Language Technology Association***: Learning to See, Integrate, and Act: Visual Grounding for Multimodal Foundation Models.
- **Sep 2026: *Siebel School Speaker Series: Machine Learning Seminar***: Learning to See, Integrate, and Act: Visual Grounding for Multimodal Foundation Models.

Projects

DARPA Environment-driven Conceptual Learning (ECOLE) Aug 2023 – Present
Funded by DARPA

DARPA In the Moment (ITM) Aug 2023 – Present
Funded by DARPA

Center for Advanced Bioenergy and Bioproducts Innovation (CABBI) Aug 2024 – Present
Funded by U.S. Department of Energy

Development of Expert Decision-making Support AI Apr 2022 – Jul 2022
Development of AI technology for explainable expert decision-making.
Funded by the Korean Government (Ministry of Science and ICT)

Contextual-Numerical Embedding for Reasoning Jul 2021 – Dec 2021
Development of context-number embedding for numerical reasoning in QA.
Funded by the Korean Government (Ministry of Science and ICT)

ExoBrain Apr 2020 – Jul 2023
Development of Knowledge Evolutionary WiseQA Platform Technology.
Funded by the Korean Government (Ministry of Science and ICT)

Smart Interaction and Context Association Apr 2020 – Mar 2021
Machine Learning for Context Association and Smart Interaction Suggestion.
Funded by the Korean Government (Ministry of Science and ICT)

AI-based National Online Petition System Apr 2021 – Dec 2021
Development of AI-based national Online Petition System for citizen deliberation.
Funded by KAISTs Exploratory Research Program

Purdue University - IITP Capstone Aug 2018 – Dec 2018
Raw radar data object detection and stealth UAV design.
Funded by the Korean Government (IITP)

Teaching

- **Aug 2020 – Jun 2021:** Teaching Assistant, Senior Data Science, Samsung SDS
- **Jun 2021 – Jul 2021:** Teaching Assistant, Text Mining, SKKU Global Business School
- **Mar 2021 – Dec 2021:** Teaching Assistant, Text Mining, KAIST
- **Mar 2019 – Jun 2019:** Teaching Assistant, Data Structures, Handong Global University

Technical Skills

Programming: Python, C, C++, Java
Frameworks: Pytorch, TensorFlow, Theano, Docker, Huggingface, OpenCV, Flutter, Firebase, DGL
Tools: SpaCy, NLTK, Seaborn, Matplotlib